

When the Treatment Talks Back: Boundary and Population Effects from Adaptive Generative Policies*

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Abstract

Conversational AI systems choose persuasive strategies adaptively, and their logs are the natural evidence about which strategies change beliefs. We study what such logs can and cannot deliver when the system’s strategy policy is known and becomes sharper, for example through a lower softmax temperature. The treatment is the choice of strategy, with the text generator held fixed. We measure the cost of learning by the deployer’s cumulative *exploration loss*: the operational value given up whenever the logged action differs from the one the deployer would have preferred.

In a one-dimensional score model we show a separation between two causal targets. The effect at the policy’s decision boundary, where the policy is indifferent, admits asymptotically valid Wald inference from logs generated by a single common temperature that shrinks with the sample size, while the cumulative exploration loss tends to zero. In the same model, two outcome laws that agree near the boundary but differ in the population average effect become statistically indistinguishable whenever the exploration loss tends to zero, so no estimator is consistent for the population effect at both. Uniformly consistent estimation of the population effect requires divergent exploration loss. In a larger model without extrapolation structure we give an information–exploration inequality valid for all estimators and adaptive designs: attaining mean squared error δ^2 requires loss of order $(\mathbb{E}\sqrt{g})^2/\delta^2$, where g is the operational value gap, and this does not decrease with deployment scale. A single deterministic common temperature makes the loss grow like $n/\log^2(n\delta^2)$, whereas uniform mixing, context-dependent temperatures, or suitable temperature mixtures keep it bounded. The results say which question adaptive logs can answer, and what it costs to answer the other one.

Keywords: adaptive experiments; limited overlap; regression discontinuity; exploration; generative AI; off-policy evaluation.

1 Introduction

Conversational artificial intelligence persuades. In randomized trials, GPT-4 shifted debate opponents’ positions more often than human debaters did (Salvi et al., 2025), and conversations

*Replication code, proofs in plain text, and all numerical results: github.com/kunhailp/when-treatment-talks-back, branch `jci-separation`. Every number in Section 7 is generated from result files by `paper/tex/make_separation_tables.py`.

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with frontier models moved policy attitudes with effects that persisted for weeks (Hackenburg et al., 2025). Establishing that a technology persuades is the first question, not the last. Regulation and replication concern narrower objects: whether presenting evidence rather than asking a question, at this point in this conversation, changes what a person believes. Systems that persuade at scale choose such strategies adaptively, and their logs are what an analyst inherits. One of the strongest levers identified by Hackenburg et al. (2025), a reward model that selects among candidate responses, is itself an adaptive selection policy.

This paper asks a precise version of the question those logs raise. Suppose the deployer’s strategy policy is known, as it is when the deployer records action probabilities, and suppose it sharpens as the deployer becomes confident, for instance by lowering a softmax temperature. Which causal effects can the logs estimate, and what must be paid to estimate the others?

The answer depends on the target. We distinguish two:

- the *boundary effect*, the effect among contexts where the policy is indifferent between strategies; and
- the *population average effect* over all contexts the deployment serves.

These answer different questions. The first is relevant to marginal changes of a policy that is already deployed; the second to whether a strategy should be used at all. Estimating one well does not substitute for estimating the other.

We price learning in the deployer’s own currency. Each time the logged strategy differs from the one the deployer’s objective prefers, the deployer gives up an operational value gap $g(h)$. The cumulative sum is the *exploration loss* R_n . This quantity is defined by the deployer’s objective, not by the human outcome under study, so it can be computed at the design stage without knowing the effects being estimated.

Contributions.

1. *Separation in one model* (Theorem 3). In a one-dimensional score model with Lipschitz mean outcomes near the decision boundary, logs generated by a single common temperature $\tau_n = n^{-\zeta}$, $\zeta \in (1/2, 1)$, give asymptotically valid Wald intervals for the boundary effect at every member of the model, while $R_n \rightarrow 0$ (Theorem 1). In the same model, whenever $R_n \rightarrow 0$, two members that coincide near the boundary but differ in the population effect have total variation distance tending to zero, so no estimator is consistent for the population effect at both; and uniform consistency for the population effect requires $\sup R_n \rightarrow \infty$.
2. *An information–exploration inequality* (Theorem 2). In the model of all bounded mean functions, for every estimator and every adaptive or non-adaptive design, worst-case mean squared error times (worst-case exploration loss plus a constant) is bounded below by $\sigma^2(\sum_k f_k \sqrt{g_k})^2$. Precision δ therefore costs exploration loss of order $(\mathbb{E}\sqrt{g})^2/\delta^2$, independent of n ; a design exploring with probability proportional to $g^{-1/2}$ attains this order (Proposition 1).
3. *What this means for temperature*. A single deterministic common temperature must be so high that exploration loss grows like $n/\log^2(n\delta^2)$ (Theorem 4). Context-dependent temperatures implement any admissible exploration design, and the optimal one is cold at the boundary and hot where the policy is confident (Proposition 2). Random mixtures of finite temperatures can also avoid the rate (Remark 1). When the value gap is misspecified, a Kantorovich bound says when gap-based allocation is worth its risk relative to uniform mixing (Lemma 1).

Scope. All results are for one decision per unit with independent contexts and a known assignment policy. Theorem 1 and Theorem 3 use a one-dimensional score, and their boundary guarantees are pointwise in the outcome law. The impossibility results hold in models without extrapolation structure: where data near the boundary identify the population effect, as under a constant effect, the separation disappears. Multi-turn conversations, estimated policies, and effects of realized text features are outside the formal results.

Relation to existing work. *Boundary estimands from algorithmic assignment.* Narita and Yata (2021) show that weights of overlap type, built from an approximate propensity score of a known algorithm, converge to a boundary estimand and require undersmoothing; this is the closest antecedent for Theorem 1. There the smoothing is an analyst-chosen ball in covariate space; here it is a temperature chosen by the system in score space, and we compute the exploration loss jointly with the inferential guarantee. Eckles et al. (2025) use known assignment noise near a cutoff for design-based inference, with a fixed noise scale and no cost accounting. Overlap weights and their target population are due to Li et al. (2018); they note that the weights peak where the propensity score is one half, but do not study a sharpening limit.

Exploration and estimation. Simchi-Levi and Wang (2023) show a regret-estimation trade-off in multi-armed bandits without contexts; Wan et al. (2022) optimize exploration for policy evaluation under a value constraint; Douglas et al. (2026) derive optimal logging policies for inverse-propensity estimation and report that softmax logging performs worse than top- k logging in simulation. The allocation $p \propto g^{-1/2}$ has the form of optimum stratified allocation with costs (Cochran, 1977); we do not claim the allocation as new. Cesa-Bianchi et al. (2017) show that Boltzmann exploration with a common schedule can fail for regret and that arm-dependent schedules repair it; our Theorem 4 and Proposition 2 are estimation analogues.

Limited overlap and off-policy evaluation. Lower bounds for off-policy value estimation (Li et al., 2015; Wang et al., 2017; Mou et al., 2023) and limited-overlap asymptotics (Khan and Tamer, 2010; Hong et al., 2020; Ma and Wang, 2020) supply the information-theoretic ingredients. Khan and Tamer (2010) observe that the average treatment effect loses identification when any region of the covariate support is removed, and derive rates for trimmed inverse weighting when propensities approach zero or one in the covariate tails; their propensities do not change with n , and there is no cost of exploration. What we add is the coupling of the information bound to the exploration loss, and the comparison of boundary and population targets on that common scale. Khan et al. (2024) bound effects without overlap under smoothness, which is the kind of extrapolation structure our impossibility excludes.

Generative treatments. Athey et al. (2026) identify effects of content features from stochastic algorithms using logged unexposed candidates and probability ratios. Their estimand averages over co-occurring pairs with trimming and does not conflict with our bounds, which concern the population effect of a strategy under outcomes observed only for the logged action. Fresh and Shin (2026) give a taxonomy of conversational estimands; we work at the level of strategy assignment. Imai and Nakamura (2026) and Nakamura and Imai (2026) study effects of realized text features under deterministic decoding; we instead hold the generation kernel fixed.

2 Setup

Generative treatments. Units $i = 1, \dots, n$ arrive with contexts $H_i \sim F$, independently. The system chooses a strategy $A_i \in \{0, 1\}$ and produces a message $M_i \sim G(\cdot \mid A_i, H_i)$ from a

fixed generation kernel G , which includes token-level sampling settings. The outcome is Y_i . The treatment is assignment of the strategy with G held fixed: potential outcomes $Y_i(a)$ are defined under the kernel intervention that sets $A_i = a$ and draws the message from G (Díaz Muñoz and van der Laan, 2012; Young et al., 2014). A label attached to a realized message after the fact is a different object, and changing G , for example by changing token temperature, changes the treatment.

Deployer. The deployer has a score gap $\Delta(h) = s_1(h) - s_0(h)$ from its own objective, a greedy strategy $a^*(h) = \mathbf{1}\{\Delta(h) > 0\}$, and an operational value gap $g(h) \geq 0$, the value its objective loses by choosing $\bar{a}(h) = 1 - a^*(h)$. We write $g = \phi(|\Delta|)$ when the gap is a function of the score.

Designs. A design specifies $P_i = \mathbb{P}(A_i = \bar{a}(H_i) \mid H_i, \mathcal{F}_{i-1})$, where $\mathcal{F}_{i-1}$ is the past data; it is non-adaptive if $P_i = p(H_i)$. A *common temperature* design sets $\mathbb{P}(A_i = 1 \mid H_i) = \text{sig}(\Delta(H_i)/\tau)$ with one deterministic $\tau = \tau_n$, so $p(h) = \text{sig}(-|\Delta(h)|/\tau_n)$, where $\text{sig}(x) = 1/(1 + e^{-x})$. The assignment probabilities are known.

Exploration loss. $R_n = \mathbb{E}[\sum_{i=1}^n g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}]$; for non-adaptive designs $R_n = n\mathbb{E}[g(H)p(H)]$.

Estimands. Let $\mu_a(h) = \mathbb{E}[Y(a) \mid H = h]$ and $c = \mu_1 - \mu_0$. The population average effect is $\theta = \mathbb{E}[c(H)]$. With a one-dimensional score $\Delta(h) = h$, the boundary effect is $\beta_0 = c(0)$. For a given temperature, the overlap-weighted effect $\beta_{\text{ov},\tau} = \mathbb{E}[\omega_\tau(H)c(H)]/\mathbb{E}[\omega_\tau(H)]$ with $\omega_\tau = p_\tau(1 - p_\tau)$ (Li et al., 2018) is a third, temperature-indexed, target; we use it only as an intermediate object.

3 Boundary effects at vanishing exploration loss

Assumption 1 (Boundary model). $\Delta(h) = h$, $a^*(h) = \mathbf{1}\{h > 0\}$, and

- (D1) H has a density $f \leq \bar{f}$, continuous at 0, with $f(0) > 0$;
- (D2) $|\mu_a| \leq B$, and μ_a is L -Lipschitz on $[-u_0, u_0]$;
- (D3) $\sigma_a^2(h) = \text{Var}(Y \mid A = a, H = h)$ is continuous at 0 with $\sigma_1^2(0) + \sigma_0^2(0) > 0$, and $\mathbb{E}[Y^4 \mid A, H] \leq M$;
- (D4) $\mathbb{P}(A = 1 \mid H = h) = \text{sig}(h/\tau_n)$ with $\tau_n \rightarrow 0$, known;
- (D5) $g = \phi(|h|) \leq \bar{G}$ with $\phi(u) \leq \bar{\kappa}u$ on $[0, u_0]$; in the strong form $\phi'(0) = \kappa > 0$.

With $w_1 = A(1 - p)$ and $w_0 = (1 - A)p$, let $\hat{\beta} = \sum w_{1i}Y_i / \sum w_{1i} - \sum w_{0i}Y_i / \sum w_{0i}$, with variance estimator $\hat{s}^2 = n^{-1} \sum \hat{\psi}_i^2$, where $\hat{\psi} = w_1(Y - \hat{m}_1)/\hat{D}_1 - w_0(Y - \hat{m}_0)/\hat{D}_0$, $\hat{D}_a = n^{-1} \sum w_{ai}$ and $\hat{m}_a$ is the weighted mean in arm a .

Theorem 1 (Boundary inference). *Under Assumption 1, if $n\tau_n \rightarrow \infty$:*

- (i) $(\hat{\beta} - \beta_{\text{ov},\tau_n})/(s_n/\sqrt{n}) \Rightarrow N(0, 1)$ with $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau_n f(0)\}(1 + o(1))$;
- (ii) $|\beta_{\text{ov},\tau_n} - \beta_0| \leq C\tau_n$ for large n ;
- (iii) $\hat{s}^2/s_n^2 \rightarrow 1$ in probability;
- (iv) if also $n\tau_n^3 \rightarrow 0$, the Wald interval $\hat{\beta} \pm z_{1-\eta/2}\hat{s}/\sqrt{n}$ has asymptotic coverage $1 - \eta$ for β_0 ;

(v) $R_n \leq n\tau_n^2 \bar{\kappa} \bar{f} \pi^2 / 6 + n\bar{G}e^{-u_0/\tau_n}$, and in the strong form $R_n = n\tau_n^2 \kappa f(0) \pi^2 / 6 (1 + o(1))$.

In particular, for $\tau_n = n^{-\zeta}$ with $\zeta \in (1/2, 1)$, the interval in (iv) is asymptotically valid and $R_n \rightarrow 0$.

The three rate conditions are $n\tau_n \rightarrow \infty$ (the boundary sample grows, $\zeta < 1$), $n\tau_n^3 \rightarrow 0$ (bias is negligible relative to the standard error, $\zeta > 1/3$), and $n\tau_n^2 \rightarrow 0$ (the loss vanishes, $\zeta > 1/2$). The error $|\hat{\beta} - \beta_0| = O_p((n\tau_n)^{-1/2})$ is achievable; we do not claim it is optimal, and it is a statement in probability rather than in mean square. Differentiability of c at zero does not improve the bias order in general: $c(h) = 1 + |h|^{3/2}$ gives bias proportional to $\tau^{3/2}$. If c is $C^{1,1}$ and f is Lipschitz near zero, the bias is $O(\tau^2)$ and condition (iv) relaxes to $n\tau_n^5 \rightarrow 0$. Proofs are in Appendix A.

4 Population effects require paid exploration

Let $\mathcal{M}(B, \sigma)$ be the model in which $Y \mid A = a, H = h$ is $N(\mu_a(h), \sigma^2)$ and μ_0, μ_1 are any measurable functions bounded by B . The model contains no structure that carries effects learned near the boundary to other contexts.

Theorem 2 (Information–exploration inequality). *Let $S_1, \dots, S_K$ be disjoint sets with $f_k = F(S_k) > 0$, each contained in $\{a^* = 0\}$ or in $\{a^* = 1\}$, and $g_k = \inf_{S_k} g$. For every design, adaptive or not, and every estimator $\hat{\theta}$,*

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \cdot \left(\sup_{\mathcal{M}(B, \sigma)} R_n + \frac{\sigma^2 \pi^2}{B^2} \sum_k g_k \right) \geq \sigma^2 \left(\sum_k f_k \sqrt{g_k} \right)^2. \quad (1)$$

Consequently, if $F(g \geq \gamma) > 0$ for some $\gamma > 0$, uniform consistency requires $\sup R_n \rightarrow \infty$; and if $\sup \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2 \rightarrow 0$, then $\liminf_n \delta_n^2 \sup R_n \geq \sigma^2 (\mathbb{E} \sqrt{g(H)})^2$.

Neither side of (1) involves n : deployment scale does not substitute for exploration when the target is the population effect. At $R_n = 0$ the bound is the finite floor $B^2 (\sum f_k \sqrt{g_k})^2 / (\pi^2 \sum g_k)$, consistent with the fact that a constant estimator has bounded error.

Proposition 1 (Achievability). *Suppose $\mathbb{E} \sqrt{g} > 0$ and $n\delta_n^2 \rightarrow \infty$. The non-adaptive design $p(h) = \min\{1/2, c_n / \sqrt{g(h)}\}$ with $c_n = \mathbb{E} \sqrt{g} / \{n\delta_n^2 / (\sigma^2 + B^2) - 4\}$ and the Horvitz–Thompson estimator give $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2$ and $R_n \leq (\sigma^2 + B^2) (\mathbb{E} \sqrt{g})^2 \delta_n^{-2} (1 + o(1))$. Uniform mixing $p \equiv (\sigma^2 + B^2) / (n\delta_n^2)$ gives $R_n = (\sigma^2 + B^2) \mathbb{E}[g] \delta_n^{-2} (1 + o(1))$.*

The rate δ^{-2} and the functional $(\mathbb{E} \sqrt{g})^2$ are therefore sharp; the constants σ^2 and $\sigma^2 + B^2$ differ because μ_a is not learned in $\mathcal{M}(B, \sigma)$. Proofs are in Appendix B.

5 The separation

Theorem 2 is proved in $\mathcal{M}(B, \sigma)$, while Theorem 1 assumes Lipschitz means near the boundary. A lower bound over a large model does not transfer to a smaller one automatically. We therefore state both sides in one model.

Let $\mathcal{M}_D$ be the set of outcome laws satisfying (D1), (D2), Gaussian noise with variance σ^2 , and (D5) with ϕ not necessarily monotone. Under Gaussian noise, (D3) holds for every member of $\mathcal{M}_D$.

Assumption 2. There are $u_0 < u_1 < u_2$ with $F([u_1, u_2]) > 0$ and $\gamma := \text{ess inf}_{h \in [u_1, u_2]} g(h) > 0$, and $B, L > 0$.

Theorem 3 (Separation). *Under Assumption 2, let $b(h) = \sin^2(\pi(h - u_1)/(u_2 - u_1))\mathbf{1}\{u_1 \leq h \leq u_2\}$, $t_{\max} = \min\{B, L(u_2 - u_1)/\pi\}$, and for $|t| \leq t_{\max}$ let $\mu^t \in \mathcal{M}_D$ have $\mu_1^t \equiv 0$ and $\mu_0^t = -tb$. All μ^t coincide on $(-\infty, u_1]$, have boundary effect $\beta_0 = 0$, and have population effect $\theta(\mu^t) = t\mathbb{E}[b(H)]$.*

(S1) *Under the common-temperature design $\tau_n = n^{-\zeta}$, $\zeta \in (1/2, 1)$, $R_n \rightarrow 0$ and, for every $\mu \in \mathcal{M}_D$, the Wald interval of Theorem 1 has asymptotic coverage $1 - \eta$ for $\beta_0(\mu)$.*

(S2) *For every design and every t , $\text{TV}(P_{\mu^0}^n, P_{\mu^t}^n) \leq R_n(\mu^0)/\gamma$. Hence if $R_n(\mu^0) \rightarrow 0$, for every $t_1 \in (0, t_{\max}]$ and every estimator, $\max_{t \in \{0, t_1\}} P_{\mu^t}(|\hat{\theta}_n - \theta(\mu^t)| \geq t_1 \mathbb{E}[b]/2) \geq (1 - R_n(\mu^0)/\gamma)/2 \rightarrow 1/2$.*

(S3) *For every design and every estimator, $\sup_{\mathcal{M}_D} \mathbb{E}(\hat{\theta} - \theta)^2 \geq (\mathbb{E}[b])^2 / \{\sup_{\mathcal{M}_D} R_n / (\sigma^2 \gamma) + \pi^2 / t_{\max}^2\}$.*

Under the design of (S1), μ^0 and μ^{t_1} are both members of $\mathcal{M}_D$ with the same boundary effect, and the boundary interval is valid at each. By (S2), no estimator of the population effect is consistent at both. By (S3), consistency uniformly over $\mathcal{M}_D$ requires unbounded exploration loss. We do not claim that the sharp constant of Theorem 2 holds in $\mathcal{M}_D$, nor any uniform guarantee on the boundary side. If a submodel lets boundary data identify θ , the perturbation family is excluded and (S2)–(S3) do not apply. Proofs are in Appendix C.

6 Temperature as an exploration device

Theorem 4 (One common temperature). *Consider common-temperature designs with one deterministic τ_n per n . Assume $|\Delta(H)|$ has a density at least $f_{\min} > 0$ on $[0, u_0]$; $\phi(u) \geq \kappa u$ on $[0, u_0]$; $G = \text{ess sup } |\Delta(H)| < \infty$; and for every $\eta' \in (0, G)$, $q_{\eta'} = \max_j \mathbb{P}(|\Delta(H)| \geq G - \eta', a^*(H) = j) > 0$. If an estimator has $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2 \rightarrow 0$, then for every $\eta' \in (0, G)$ and all large n*

$$R_n \geq \kappa f_{\min} c_0 n \min \left\{ u_0^2, \frac{(G - \eta')^2}{\log^2 \{2n\delta_n^2 / (q_{\eta'} \sigma^2)\}} \right\}, \quad c_0 = \int_0^1 v \text{sig}(-v) dv.$$

Compared with Proposition 1, the relative cost grows like $n\delta_n^2 / \log^2(n\delta_n^2)$. With finitely many score values $0 < \Delta_{\min} < \Delta_{\max}$ and fixed, sufficiently small δ , the loss grows like $n^{1 - \Delta_{\min}/\Delta_{\max}}$. The reason is not that a common temperature explores in the wrong place: the optimal design also explores most near the boundary, where exploration is cheap. It is that the exploration probability of a common temperature decays exponentially in $|\Delta|$, so observations far from the boundary become too rare unless the temperature is raised everywhere, which is expensive near the boundary.

Remark 1 (Mixtures of temperatures). Theorem 4 does not extend to random mixtures of finite temperatures. With $|\Delta| = |H|$, $H \sim U(-1, 1)$, $g = |H|$, mixing temperature 1 with weight $w_n = \mathbb{E}[1/\text{sig}(-|H|)]/(n\delta^2)$ and temperature $1/n$ otherwise keeps the design variance at δ^2 with bounded loss (46.36 at $\delta = 0.1$ for $n = 10^4, 10^5, 10^6$). The lesson concerns a single common temperature, not temperature as such.

δ	design	RMSE	coverage	MCSE	explored	loss R_n
0.10	uniform mixing	0.098	0.950	0.011	100	49.9
0.10	$p \propto g^{-1/2}$ (= context temp.)	0.103	0.932	0.013	134	44.5
0.10	common temperature	0.097	0.965	0.009	7607	988.5
0.05	uniform mixing	0.052	0.955	0.010	400	200.3
0.05	$p \propto g^{-1/2}$ (= context temp.)	0.053	0.940	0.012	532	178.0
0.05	common temperature	0.050	0.948	0.011	9191	1440.1

Table 1: Exploration designs at $n = 10^5$, 400 replications with common random numbers. Coverage is of the population effect by Wald intervals; MCSE uses the observed coverage. “Explored” is the mean number of off-greedy actions.

Proposition 2 (Context temperatures). *For $\Delta(h) \neq 0$ and $p(h) \in (0, 1/2)$, the temperature $\tau(h) = |\Delta(h)|/\log\{(1 - p(h))/p(h)\}$ gives off-greedy probability $p(h)$; $p = 1/2$ requires $\tau = \infty$, and on $\{\Delta = 0\}$ only $p = 1/2$ is implementable. The design of Proposition 1 satisfies these conditions, and its temperature satisfies $\tau^*(h)\log(1/c_n) \rightarrow |\Delta(h)|$ for each h with $\Delta(h) \neq 0$: cold at the boundary, hot where the policy is confident.*

Lemma 1 (Misspecified gaps). *Allocate with $\hat{g}$ in place of g at the same design variance, with $\hat{g}/g \in [a, b]$ on $\{g > 0\}$ and $\rho = b/a$. Then the loss ratio is at most $K(\rho) = (1 + \sqrt{\rho})^2/(4\sqrt{\rho})$. The bound is attained when the measure $w \propto \sqrt{g}dF$ has a set of mass exactly one half, for example when it is non-atomic.*

A common rescaling of $\hat{g}$ changes nothing; only relative misspecification across contexts matters. Gap-based allocation beats uniform mixing for every admissible $\hat{g}$ if $\mathbb{E}[g]/(\mathbb{E}\sqrt{g})^2 > K(\rho)$; the condition is also necessary when the bound is attained, but not in general. Proofs are in Appendix D.

7 Simulations

All simulations use $H \sim U(-1, 1)$, $\Delta(h) = h$, $g(h) = |h|$, $\mu_0(h) = h$, $c(h) = 1 + |h|$ (population effect 1.5, boundary effect 1), and $N(0, 1)$ noise.

Exploration designs. At $n = 10^5$ and design variance $\delta^2 = 0.01$, the closed-form exploration loss is 50.0 for uniform mixing, 44.4 for $p \propto g^{-1/2}$, and 988.2 for a common temperature ($\tau = 0.110$), a ratio of 19.8 to uniform mixing. Table 1 runs the augmented inverse-propensity estimator with a linear working model on common random numbers across designs. The context-temperature implementation of $p \propto g^{-1/2}$ produces identical rows and is not listed separately. In a two-context example the bound (1) evaluates to 51.9 against the asymptotic constant 52.4, and the common-temperature loss has local growth exponent 0.808 against the predicted 0.8.

Boundary inference. Table 2 reports the boundary estimator of Theorem 1. Exploration loss matches the constant in Theorem 1(v). With $\zeta = 0.6$ coverage is close to nominal; at $n = 10^6$, 5,000 replications give 0.9458 (MCSE 0.0032), with the estimated standard error at 0.993 of the Monte Carlo standard deviation. With a very small boundary sample the interval under-covers: 0.887 at $n\tau_n \approx 10$. The guarantee of Theorem 1 is asymptotic.

ζ	n	$n\tau_n$	reps	RMSE	coverage of β_0	MCSE	loss R_n (theory)
0.60	10,000	39.8	1,000	0.215	0.949	0.007	0.1290 (0.1304)
0.60	100,000	100.0	500	0.141	0.944	0.010	0.0821 (0.0822)
0.60	1,000,000	251.2	200	0.082	0.970	0.012	0.0513 (0.0519)
0.75	10,000	10.0	1,000	0.464	0.887	0.010	0.0081 (0.0082)
0.75	100,000	17.8	500	0.339	0.924	0.012	0.0026 (0.0026)
0.75	1,000,000	31.6	200	0.247	0.930	0.018	0.0008 (0.0008)
0.60	1,000,000	251.2	5,000	–	0.946	0.003	–

Table 2: Boundary inference under common temperature $\tau_n = n^{-\zeta}$. Coverage is of $\beta_0 = c(0)$. Loss is the Monte Carlo mean with the Theorem 1(v) constant in parentheses. The last row is a separate 5,000-replication run.

8 Discussion

The results separate two uses of adaptive logs. If the question is how outcomes respond where the deployed policy is indifferent, logs from a sharpening policy are informative, and increasing scale can replace exploration. If the question is the population average effect of a strategy, the logs are informative only to the extent the deployer has paid exploration loss away from the boundary, and increasing scale cannot replace it. Reporting an effect estimated from adaptive logs therefore requires saying which of the two quantities it is.

For design, the relevant choice is not whether to use temperature but how. With one temperature shared by all contexts, the exploration probability decays exponentially in the policy’s confidence, so reaching confident contexts forces a temperature that is costly near the boundary; uniform mixing, context-dependent temperatures, or temperature mixtures with a vanishing weight on a moderate temperature avoid this. Among designs that avoid it, allocating by the value gap saves further loss when gaps are heterogeneous and reasonably well specified (Lemma 1); in our example the saving is about ten percent.

Limitations. The policy is known and assignments are logged; the setting is a single decision per unit with independent contexts; the boundary results are one-dimensional and pointwise; the population impossibility excludes models in which boundary data identify the population effect; and the treatment is strategy assignment under a fixed generator, not the realized text. Extending the separation to multi-turn conversations requires defining the reference policy for later turns and the induced distribution of histories, which we leave open. Appendix E reports descriptive re-query evidence on how concentrated strategy choices of open-weights models are; it does not measure any deployed policy’s assignment probabilities.

Data and code availability. Proofs in plain text, simulation code, result files with arguments, seeds and code commits, and the scripts that generate every number in Section 7 are in the repository. The re-query analysis in Appendix E uses the public DebateGPT release (Salvi et al., 2025); the file of extracted contexts includes participants’ opening-argument text from that release. The fidelity audit uses released rows of Hackenburg et al. (2025), which do not include conversation transcripts.

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A Proof of Theorem 1

Write $p = \text{sig}(H/\tau)$, $\omega = p(1 - p) = K(H/\tau)$ with $K(u) = e^u/(1 + e^u)^2$, and $m_\tau = \mathbb{E}\omega$. We use

$$\int K = 1, \quad \int |u|K(u) du = 2 \log 2, \quad \int K(u)\text{sig}(u) du = \frac{1}{2}, \quad K(u) \leq e^{-|u|}, \quad \int_0^\infty v \text{sig}(-v) dv = \frac{\pi^2}{12}.$$

Step 1 (kernel integrals). For q bounded and continuous at 0, $j \geq 1$ and $r, s \geq 0$, substituting $h = \tau u$ and using dominated convergence with dominating function $\|q\|_\infty \bar{f}K(u)$,

$$\mathbb{E}[K(H/\tau)^j (1 - p)^r p^s q(H)] = \tau f(0)q(0) \int K^j \text{sig}(-u)^r \text{sig}(u)^s du + o(\tau).$$

In particular $m_\tau = \tau f(0)(1 + o(1))$.

Step 2 (bias). $\beta_{\text{ov},\tau} - \beta_0 = \mathbb{E}[\omega\{c(H) - c(0)\}]/m_\tau$. On $|H| \leq u_0$, $|c(h) - c(0)| \leq 2L|h|$ and $\mathbb{E}[\omega|H|] \leq \bar{f}\tau^2 \int |u|K = 2 \log 2 \bar{f}\tau^2$. On $|H| > u_0$, $\omega \leq e^{-u_0/\tau}$ and $|c(h) - c(0)| \leq 4B$. Hence $|\beta_{\text{ov},\tau} - \beta_0| \leq (4 \log 2 L \bar{f}\tau^2 + 4Be^{-u_0/\tau})/m_\tau = (4 \log 2 L \bar{f}/f(0))\tau(1 + o(1))$.

Step 3 (linearization). Let $\bar{\mu}_a = \mathbb{E}[w_a \mu_a]/m_\tau$, so $\bar{\mu}_1 - \bar{\mu}_0 = \beta_{\text{ov},\tau}$. Since $\mathbb{E}[w_1 | H] = \omega$ and $\mathbb{E}[w_1 Y | H] = \omega \mu_1$, $\mathbb{E}[w_1(Y - \bar{\mu}_1)] = 0$, and likewise for arm 0. Define $T_a = n^{-1} \sum_i w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$ and $\xi_i = \{w_{1i}(Y_i - \bar{\mu}_1) - w_{0i}(Y_i - \bar{\mu}_0)\}/m_\tau$. Then $\hat{m}_a - \bar{\mu}_a = (m_\tau/\hat{D}_a)T_a$ and $\mathbb{E}\xi = 0$. Because $w_1 w_0 = 0$ and $\mathbb{E}[A(1-p)^2 | H] = \omega(1-p)$,

$$s_n^2 := \text{Var}\xi = m_\tau^{-2} \mathbb{E}[\omega\{(1-p)(\sigma_1^2 + (\mu_1 - \bar{\mu}_1)^2) + p(\sigma_0^2 + (\mu_0 - \bar{\mu}_0)^2)\}].$$

By Step 1, $\bar{\mu}_a \rightarrow \mu_a(0)$; the terms in $(\mu_a - \bar{\mu}_a)^2$ are $o(\tau)$ by dominated convergence; and the variance terms equal $\tau f(0)\sigma_a^2(0)/2 + o(\tau)$. Hence $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau f(0)\}(1 + o(1))$.

Step 4 (central limit theorem). Since $0 \leq w_a \leq 1$, $w_a^4 \leq w_a$, and $\mathbb{E}[(Y - \bar{\mu}_a)^4 | A, H] \leq C(M, B)$, we get $\mathbb{E}[w_a^4(Y - \bar{\mu}_a)^4] \leq C m_\tau$ and $\mathbb{E}\xi^4 \leq C' m_\tau^{-3}$. The Lyapunov ratio $n^{-1} \mathbb{E}\xi^4/s_n^4 \asymp (n\tau)^{-1} \rightarrow 0$, so $n^{-1/2} \sum \xi_i/s_n \Rightarrow N(0, 1)$. Also $\text{Var}(\hat{D}_a)/m_\tau^2 \leq 1/(nm_\tau) \rightarrow 0$, so $\hat{D}_a/m_\tau \rightarrow 1$ in probability and $m_\tau/\hat{D}_a - 1 = O_p((n\tau)^{-1/2})$. Now

$$\hat{\beta} - \beta_{\text{ov},\tau} = (T_1 - T_0) + (m_\tau/\hat{D}_1 - 1)T_1 - (m_\tau/\hat{D}_0 - 1)T_0,$$

with $T_1 - T_0 = n^{-1} \sum \xi_i$ and $T_a = O_p((n\tau)^{-1/2})$. The remainder is $O_p((n\tau)^{-1}) = o_p(s_n/\sqrt{n})$, and Slutsky's lemma gives (i).

Step 5 (variance estimator). Write $\xi_{ai} = w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$. First, $n^{-1} \sum \xi_i^2/s_n^2$ has mean one and variance at most $\mathbb{E}\xi^4/(ns_n^4) \asymp (n\tau)^{-1}$. Second,

$$\hat{\psi}_{1i} - \xi_{1i} = w_{1i}(Y_i - \bar{\mu}_1)(1/\hat{D}_1 - 1/m_\tau) + w_{1i}(\bar{\mu}_1 - \hat{m}_1)/\hat{D}_1.$$

The mean square of the first term is $(n^{-1} \sum \xi_{1i}^2)(m_\tau/\hat{D}_1 - 1)^2 = O_p(s_n^2)O_p((n\tau)^{-1})$. For the second, $w_1^2 \leq w_1$ gives mean square at most $(\hat{m}_1 - \bar{\mu}_1)^2/\hat{D}_1$; by Step 4 and $\hat{D}_1 = \tau f(0)(1 + o_p(1))$ this is $O_p(1/(n\tau^2))$, which is $O_p((n\tau)^{-1})$ relative to $s_n^2 \asymp \tau^{-1}$. Arm 0 is identical. By Minkowski's and the Cauchy-Schwarz inequalities, $|n^{-1} \sum \hat{\psi}_i^2 - n^{-1} \sum \xi_i^2| = o_p(s_n^2)$, which gives (iii).

Step 6 (centering at β_0). $(\beta_{\text{ov},\tau} - \beta_0)/(s_n/\sqrt{n}) = O(\tau)O((n\tau)^{1/2}) = O((n\tau^3)^{1/2})$. With (i), (iii) and Slutsky's lemma this gives (iv).

Step 7 (exploration loss). The off-greedy probability is $\text{sig}(-|h|/\tau)$. On $|h| \leq u_0$, $n\mathbb{E}[\phi(|H|)\text{sig}(-|H|/\tau)] \leq n\bar{\kappa}\bar{f} \cdot 2\tau^2 \int_0^\infty v \text{sig}(-v)dv = n\bar{\kappa}\bar{f}\tau^2\pi^2/6$, and the rest is at most $n\bar{G}e^{-u_0/\tau}$. In the strong form, on $v \leq u_0/\tau$, $\phi(\tau v)/\tau \rightarrow \kappa v$ and $f(\tau v) \rightarrow f(0)$ with dominating function $\bar{\kappa}v\bar{f}\text{sig}(-v)$, which gives the asymptotic equality. For $\tau_n = n^{-\zeta}$: $n\tau_n \rightarrow \infty$ iff $\zeta < 1$; $n\tau_n^3 \rightarrow 0$ iff $\zeta > 1/3$; $n\tau_n^2 \rightarrow 0$ iff $\zeta > 1/2$. $\square$

Smoother effects. If $|c(h) - c(0) - c'(0)h| \leq M_2 h^2$ and f is L_f -Lipschitz near 0, then $\mathbb{E}[\omega H] = \tau^2 \int u K(u) \{f(\tau u) - f(0)\} du = O(\tau^3)$ up to an exponentially small tail, and the quadratic remainder contributes at most $M_2 \bar{f} \tau^3 \int u^2 K(u)/m_\tau$; the bias is $O(\tau^2)$. Differentiability of c alone is not enough: $c(h) = 1 + |h|^{3/2}$ gives bias $\tau^{3/2} \int |u|^{3/2} K(u) du (1 + o(1))$.

B Proofs for Section 4

Proof of Theorem 2. *Submodel.* For $\vartheta \in [-B, B]^K$ set $\mu_{a^*(h)}^\vartheta(h) = 0$, and on S_k set $\mu_{\bar{a}(h)}^\vartheta(h) = s_k \vartheta_k$ with $s_k = +1$ if $\bar{a} = 1$ on S_k and $s_k = -1$ otherwise; elsewhere $\mu_{\bar{a}(h)}^\vartheta(h) = 0$. Then $\mu^\vartheta \in \mathcal{M}(B, \sigma)$ and $\theta(\mu^\vartheta) = \sum_k f_k \vartheta_k$.

Information. The sample density is $\prod_i dF(H_i) \kappa_i(A_i | H_i, \mathcal{F}_{i-1}) \varphi_\sigma(Y_i - \mu_{A_i}^\vartheta(H_i))$, where the assignment kernels κ_i are known functions of observed data. The score for ϑ_k is $\sum_i \mathbf{1}\{H_i \in S_k, A_i = \bar{a}(H_i)\} s_k (Y_i - s_k \vartheta_k) / \sigma^2$. Its summands are martingale differences, so the Fisher information matrix is diagonal with $I_k(\vartheta) = \mathbb{E}_\vartheta N_k / \sigma^2$, where $N_k = \sum_i \mathbf{1}\{H_i \in S_k, A_i = \bar{a}(H_i)\}$.

van Trees. Take the product prior with marginal density $B^{-1} \cos^2(\pi x / (2B))$ on $[-B, B]$, whose Fisher information is $J = \pi^2 / B^2$. With $\bar{N}_k = \int \mathbb{E}_\vartheta N_k \lambda(d\vartheta)$, the multivariate van Trees inequality (Gill and Levit, 1995), maximized over the direction vector, gives

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \sum_k \frac{f_k^2}{\bar{N}_k / \sigma^2 + J} = \sum_k \frac{f_k \sigma^2}{a_k}, \quad a_k = \frac{\bar{N}_k + \sigma^2 J}{f_k}.$$

Cauchy–Schwarz and loss. Writing $\sigma \sum_k f_k \sqrt{g_k} = \sum_k \sqrt{f_k g_k a_k} \cdot \sigma \sqrt{f_k / a_k}$,

$$\sigma^2 \left(\sum_k f_k \sqrt{g_k} \right)^2 \leq \left(\sum_k f_k g_k a_k \right) \left(\sum_k f_k \sigma^2 / a_k \right).$$

Here $\sum_k f_k g_k a_k = \sum_k g_k \bar{N}_k + \sigma^2 J \sum_k g_k$, and since $g \geq g_k$ on S_k , $\sum_k g_k N_k \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$, so $\sum_k g_k \bar{N}_k \leq \sup R_n$. This proves (1).

Consequences. With $K = 1$ and $S_1 = \{g \geq \gamma\} \cap \{a^* = j\}$ of positive mass, (1) gives $\sup R_n \geq g_1 \{\sigma^2 f_1^2 / \sup \mathbb{E}(\hat{\theta} - \theta)^2 - \sigma^2 \pi^2 / B^2\} \rightarrow \infty$. For the constant, fix a partition into level sets of g intersected with the two greedy sides; (1) gives $\liminf \delta_n^2 \sup R_n \geq \sigma^2 (\sum f_k \sqrt{g_k})^2$, and lower sums $\sum F(S) \inf_S \sqrt{g}$ increase to $\mathbb{E} \sqrt{g}$ as the level mesh shrinks. $\square$

Proof of Proposition 1. The Horvitz–Thompson estimator is unbiased with variance at most $n^{-1}(\sigma^2 + B^2) \mathbb{E}[1/e_1 + 1/(1 - e_1)] \leq n^{-1}(\sigma^2 + B^2)(\mathbb{E}[1/p] + 2)$, where $e_1 = \mathbb{P}(A = 1 | H)$. Pointwise $1/p = \max\{2, \sqrt{g}/c_n\} \leq \sqrt{g}/c_n + 2$, and the choice of c_n gives $\mathbb{E}[1/p] + 2 \leq n \delta_n^2 / (\sigma^2 + B^2)$. The loss is $n \mathbb{E}[gp] \leq n c_n \mathbb{E} \sqrt{g}$. The uniform design is analogous. $\square$

C Proof of Theorem 3

The function b satisfies $0 \leq b \leq 1$, is $\pi/(u_2 - u_1)$ -Lipschitz, vanishes on $(-\infty, u_1]$, and has $\mathbb{E}b(H) > 0$ because H has a density and $F([u_1, u_2]) > 0$. Hence $|tb| \leq B$ and tb is L -Lipschitz for $|t| \leq t_{\max}$, so $\mu^t \in \mathcal{M}_D$; $c^t = tb$, $\beta_0(\mu^t) = 0$, and $\theta(\mu^t) = t \mathbb{E}b(H)$. Every perturbed observation is off-greedy, since $a^* = 1$ on $(0, \infty)$, with $H \in [u_1, u_2]$. Let $N = \sum_i \mathbf{1}\{A_i = 0, H_i \in [u_1, u_2]\}$.

(S1) is Theorem 1, because every member of $\mathcal{M}_D$ satisfies Assumption 1.

(S2). The sample densities under μ^0 and μ^t coincide on $E = \{N = 0\}$, so $P_0(E) = P_t(E)$, and for any event A , $|P_0(A) - P_t(A)| = |P_0(A \cap E^c) - P_t(A \cap E^c)| \leq \max\{P_0(E^c), P_t(E^c)\} = P_0(E^c)$. Thus $\text{TV} \leq P_0(N \geq 1) \leq \mathbb{E}_0 N$. Because $g \geq \gamma$ F -almost surely on $[u_1, u_2]$ and $H_i \sim F$, $\gamma N \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$ almost surely, so $\mathbb{E}_0 N \leq R_n(\mu^0) / \gamma$. Le Cam’s two-point argument with separation $t_1 \mathbb{E}b$ gives the displayed bound.

(S3). The score of $t \mapsto \mu^t$ is $\sum_i \mathbf{1}\{A_i = 0\} (-b(H_i))(Y_i + tb(H_i)) / \sigma^2$, with information $I(t) = \mathbb{E}_t \sum_i \mathbf{1}\{A_i = 0\} b(H_i)^2 / \sigma^2 \leq \mathbb{E}_t N / \sigma^2 \leq R_n(\mu^t) / (\gamma \sigma^2)$. With the prior $t_{\max}^{-1} \cos^2(\pi t / (2t_{\max}))$ on

$[-t_{\max}, t_{\max}]$, whose information is $\pi^2/t_{\max}^2$, the van Trees inequality for $\psi(t) = t\mathbb{E}b$ gives the bound. $\square$

D Proofs for Section 6

Proof of Theorem 4. Take the single cell $S = \{|\Delta| \geq G - \eta', a^* = j\}$, with j attaining $q = q_{\eta'}$. There $p \leq e^{-(G-\eta')/\tau}$, so $\bar{N} \leq nqe^{-(G-\eta')/\tau}$, and the van Trees step of Theorem 2 with $K = 1$ gives $\delta_n^2 \geq q^2\sigma^2/(nqe^{-(G-\eta')/\tau} + \sigma^2\pi^2/B^2)$. Once $\delta_n^2 \leq q^2B^2/(2\pi^2)$, this forces $nqe^{-(G-\eta')/\tau} \geq q^2\sigma^2/(2\delta_n^2)$, that is, $\tau_n \geq (G - \eta')/\log\{2n\delta_n^2/(q\sigma^2)\}$. On the other hand $R_n = n\mathbb{E}[\phi(|\Delta|)\text{sig}(-|\Delta|/\tau)] \geq n\kappa f_{\min}\tau^2 \int_0^{u_0/\tau} v\text{sig}(-v)dv \geq n\kappa f_{\min}c_0 \min\{\tau, u_0\}^2$. Combining the two displays proves the claim. With finitely many score values, the same argument gives $\tau \geq \Delta_{\max}/\log(Cn\delta^2)$ for δ small enough, and the $\Delta_{\min}$ cell contributes $nf\phi(\Delta_{\min})e^{-\Delta_{\min}/\tau}/2$. $\square$

Remark 1. With $q = \text{sig}(-u)$ and $p_n = w_nq + (1-w_n)\text{sig}(-nu)$, $p_n \geq w_nq$ gives $\mathbb{E}[1/p_n]/n \leq \delta^2$ for $w_n = \mathbb{E}[1/q]/(n\delta^2)$, and $n\mathbb{E}[up_n] \leq \mathbb{E}[1/q]\mathbb{E}[uq]/\delta^2 + O(1/n)$. With $w_n = \mathbb{E}[1/q]/\{n\delta_n^2/(\sigma^2 + B^2) - 2\}$ the Horvitz–Thompson risk is at most δ_n^2 uniformly over $\mathcal{M}(B, \sigma)$.

Proof of Proposition 2. Solving $\text{sig}(-|\Delta|/\tau) = p$ gives the formula. For $p^* = c_n/\sqrt{\phi(|\Delta|)} < 1/2$, $\log\{(1-p^*)/p^*\} = \log(1/c_n) + \frac{1}{2}\log\phi(|\Delta|) + \log(1-p^*)$, and $c_n \rightarrow 0$. On $\{\Delta = 0\}$, $g = \phi(0) = 0$, so $p^* = 1/2$. $\square$

Proof of Lemma 1. With $p = c/\sqrt{\hat{g}}$ and $\mathbb{E}[1/p] = T$, the loss is $n\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/T$, against $n(\mathbb{E}\sqrt{g})^2/T$ for $\hat{g} = g$. With $s = (\hat{g}/g)^{1/2} \in [\sqrt{a}, \sqrt{b}]$ and the probability measure $w \propto \sqrt{g}dF$, the ratio is $\mathbb{E}_w[s]\mathbb{E}_w[1/s]$, which Kantorovich’s inequality bounds by $(\sqrt{a} + \sqrt{b})^2/(4\sqrt{ab}) = K(b/a)$. If w has a set of mass one half, setting $s = \sqrt{a}$ there and $s = \sqrt{b}$ elsewhere attains the bound. With atoms the bound need not be attained: for $F = (0.99, 0.01)$, $g = (1, 100)$ and $\rho = 36$, uniform mixing costs 1.675 times the optimum, below $K(36) = 2.042$, yet every admissible $\hat{g}$ costs at most 1.347 times the optimum. $\square$

E Descriptive re-query evidence

We re-queried three instruction-tuned open-weights models 20 times on each of 200 debate contexts and labeled each generation with an LLM judge. Per-context label distributions were highly concentrated. We do not read this as a measurement of any deployed policy. The generations are lexically diverse within a context, so any concentration arises only after projection onto labels; agreement between two judges was low ($\kappa = 0.13$); and the prompt asked for a rebuttal while the modal label was “rebuttal”, which makes that label close to tautological. With 20 draws a strategy with probability 0.02 goes unobserved about two thirds of the time. The evidence indicates at most that strategy labels of instruction-tuned models can be concentrated enough for the regime studied here to matter.